AWS CloudTrail for Auditing

Explore AWS CloudTrail to understand how to audit and monitor API activity within your AWS account. Learn about CloudTrail's event types, event history, trails, Lake, and Insights. This lesson helps you gain insights into securing, troubleshooting, and managing your AWS resources effectively.

AWS CloudTrail is a service that records AWS account activity by logging API calls made through the console, SDKs, CLI, or other services. It provides a history of events such as user logins, resource changes, and service access, helping with security auditing, compliance, and troubleshooting.

CloudTrail events

Any activity within our AWS account is recorded as a **CloudTrail event**. CloudTrail event history is automatically enabled by default when we create an AWS account, so we do not need to perform any steps manually.

CloudTrail events are of the following types:

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- **Management events:** Management events represent actions performed on resources within our AWS account. For example, launching an EC2 instance

(`RunInstances`) is considered a management event.

- **Data events:** Data events represent object-level actions performed within specific AWS services, like Amazon S3 or AWS Lambda. For example, uploading an object (`PutObject`) or deleting an object (`DeleteObject`) in an S3 bucket are considered data events, as these are actions related to the S3 object. In retrospect, creating an S3 bucket (`CreateBucket`) itself is a management event, as we're not performing any action within a resource, but actually, the action is to create a resource.

> **Disclaimer:** Please note that CloudTrail does not log data in real time, and we should expect some delay when the event is logged.

CloudTrail services

CloudTrail is an expansive service that we can break into the following sub-services.

CloudTrail event history

CloudTrail event history is a service within CloudTrail that allows us to view, search, and download our AWS account's immutable historical activity data for the last 90 days. It is useful for ad-hoc investigations and quick lookups into recent account activities. By default, CloudTrail event history only logs management events.

CloudTrail allows developers to look for events using the AWS CloudTrail `lookup-events` command. For example, we can look for the event history of an EC2 instance through its ID using the command given below:

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main.sh Shell

```
aws cloudtrail lookup-events \
  --lookup-attributes AttributeKey=ResourceName,AttributeValue=i-0abc
```

> **Note:** CloudTrail Event History is free to use and allows us to view and fetch the event history of our AWS account for the last 90 days at no additional cost.

CloudTrail trail

CloudTrail trail allows us to store event history for more than 90 days. We can create a trail to store management events, data events, insight events, or all of them in one place.

We can configure a trail to log events to long-term storage, like:

- An S3 bucket for long-term storage
- A CloudWatch log group for real-time monitoring and alarms
- A Lambda function for automations and data processing

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Storing long-term CloudTrail logs

We can also customize trails to store events within specific or all regions, or even all accounts within an organization account.

To understand how trials work, let's consider a simple example. We want to create a trail to log data events for Lambda functions. For that, we first need to create a trail that logs data events using the command given below:

main.sh Shell

```
aws cloudtrail create-trail \
  --name data-events-trail \
  --s3-bucket-name my-cloudtrail-logs-bucket \
  --is-multi-region-trail
```

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Command to create a data events trail

Next, we need to start the logging for the trail using the command given below:

main.sh Shell

```
aws cloudtrail start-logging --name data-events-trail
```

Once the trail starts logging, we need to configure an event selector for the trail to specify what types of events should be logged. For example, if we want to log all read and write data events for a specific Lambda function, and also include standard management events like function creation or deletion, we'll use the following command:

main.sh Shell

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```
aws cloudtrail put-event-selectors \
  --trail-name data-events-trail \
  --event-selectors '[
    {
      "ReadWriteType": "All",
```

```
        "IncludeManagementEvents": true,
        "DataResources": [
          {
            "Type": "AWS::Lambda::Function",
            "Values": ["arn:aws:lambda:us-east-1:123456789012:function:r
          }
        ]
```

Command to put an event selector for the trail

Note: The pricing for CloudTrail trails is based on whether we've enabled any and set up any management or data events or integrated it with S3 and SNS services, which have their own separate costs. Depending on our usage, the cost for CloudTrail trails is variable.

CloudTrail Lake

CloudTrail Lake is a managed service within CloudTrail that offers features like security lake and auditing support to provide a unified view of all our AWS CloudTrail event data. It supports monitoring activity within a single region, across all regions, or across all accounts in an AWS Organization.

CloudTrail Lake allows us to aggregate, immutably store, and query activity logs for operational and security insights of our AWS cloud. Compared to CloudTrail event history, CloudTrail Lake is designed for more complex queries, long-term event data storage, and analysis of compliance and security incidents over extended periods.

CloudTrail pricing depends on the configuration we choose. Costs vary depending on whether we enable trails, management events, or data events and whet integrate CloudTrail with services like Amazon S3 or Amazon SNS, which have separate charges. Depending on how extensively we use these features, the overall cost can range from low to high.

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> CloudTrail Trail is a delivery service that sends raw files and logs to an S3 bucket for long-term storage. On the other hand, CloudTrail Lake is an all-in-one managed service with an advanced SQL query engine for advanced analysis.

CloudTrail Insights

CloudTrail Insights is an advanced service within CloudTrail that leverages AI and ML models to detect and notify us of any unusual activity within our AWS account. It monitors our activity regularly and constructs a baseline based on it. Using this baseline, it can identify unusual patterns or activity in our AWS account, like spikes in the usual activity, rarely or not at all used services being utilized, or provisioning of more costly resources than normal.

Hence, CloudTrail Insights can be helpful in identifying security breaches like compromised access credentials or insider threats. It also provides timely notifications to investigate the threat and mitigate the damage already done.

> Cloud Trail also logs another type of event called **CloudTrail Insight events**. They represent any actions that the CloudTrail Insight service considers unusual within our AWS account. For example, if our daily activity includes launching and deleting ~5 EC2 instances in our AWS account, which CloudTrail Insight sets as the baseline, the launch of ~50 EC2 instances would

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> be considered a CloudTrail insight event, as that's considered an unusual activity.

Conclusion

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